



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

TECHNICAL MATHEMATICS P2/TEGNIJSE WISKUNDE V2

MAY/JUNE 2024

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

CODE/ KODE	EXPLANATION/VERDUIDELIKING
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Volgehoue akkuraatheid
I	Identity/Identiteit
M	Method/Metode
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for omitting units/Geen penalisering vir eenhede weggelaat nie
R	Rounding/Afronding
RE	Reason/Rede
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule
ST/RE	Statement with reason/Bewering met rede
F	Correct formula/Korrekte formule

**These marking guidelines consist of 20 pages.
Hierdie nasienriglyne bestaan uit 20 bladsye.**

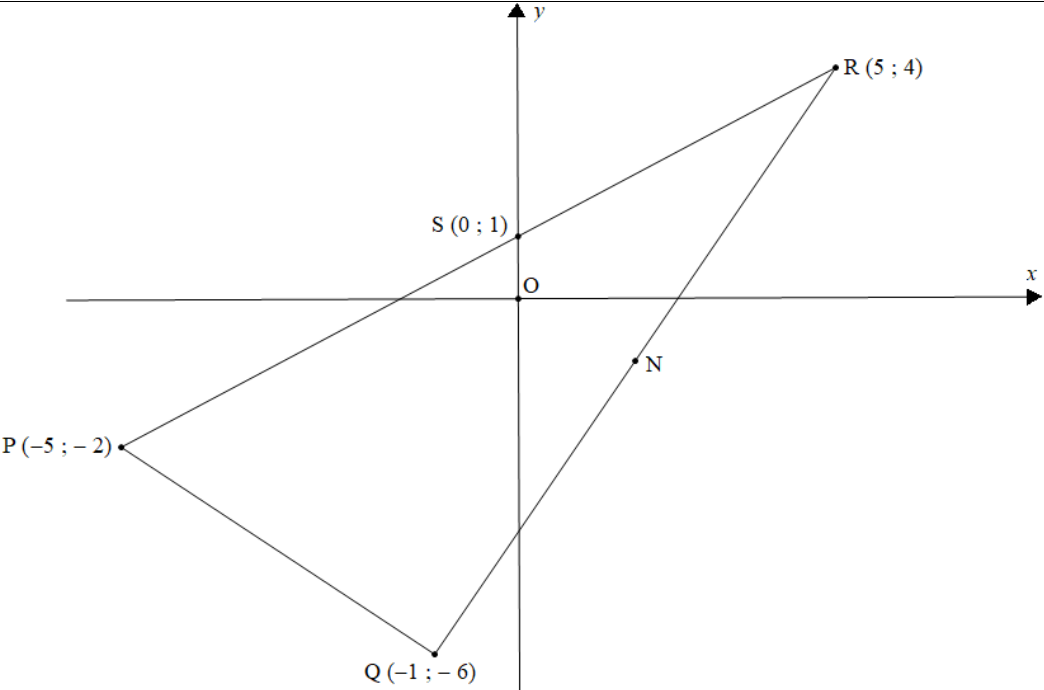
NOTE:

- If a candidate answers a question **TWICE**, only mark the **FIRST** attempt.
- The method of Consistent Accuracy marking must be applied in all aspects of the marking guideline where indicated with the marking code **CA**.

LET WEL:

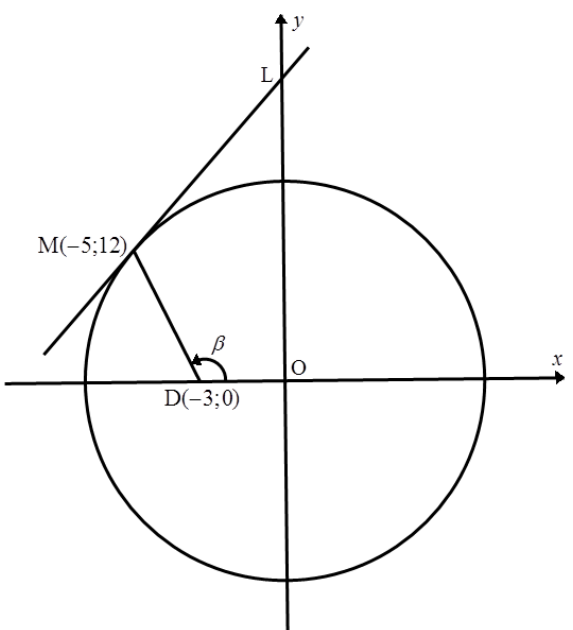
- Indien 'n kandidaat 'n vraag **TWEE** keer beantwoord, sien slegs die **EERSTE** poging na.
- Die metode van Volgehoue Akkuraatheid-nasien moet waar moontlik op alle aspekte van die nasienriglyne toegepas word soos aangedui deur die nasienkode **CA**.

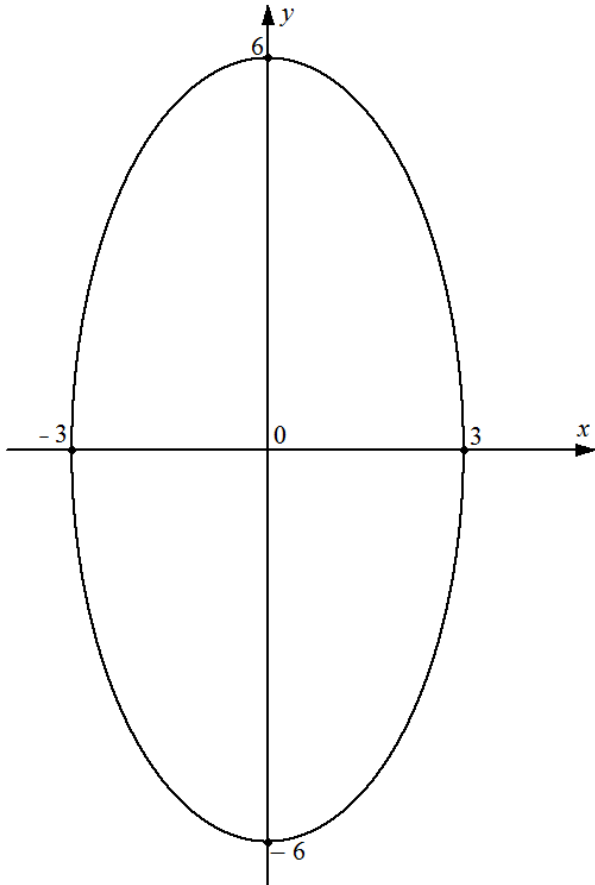
QUESTION/VRAAG 1

			
1.1.1	$m_{PQ} = \frac{-2 - (-6)}{-5 - (-1)}$ $= -1$	✓ SF ✓ gradient/gradient	A CA (2)
1.1.2	$N\left(\frac{x_R + x_Q}{2}; \frac{y_R + y_Q}{2}\right)$ $N\left(\frac{5 + (-1)}{2}; \frac{4 + (-6)}{2}\right)$ $N(2; -1)$ OR/OF $x_N = \frac{x_1 + x_2}{2}; y_N = \frac{y_1 + y_2}{2}$ $x_N = \frac{5 + (-1)}{2}, y_N = \frac{4 + (-6)}{2}$ $N(2; -1)$	✓ x-value/waarde ✓ y-value /waarde OR/OF ✓ x-value/waarde ✓ y-value /waarde AO: full marks/ volpunte	A A A A (2)

1.1.3	$y = -1x + c$ $4 = -1(5) + c$ $\therefore c = 9$ $\therefore y = -x + 9$ OR/OF $y - y_1 = m(x - x_1)$ $y - 4 = -1(x - 5)$ $y - 4 = -x + 5$ $\therefore y = -x + 9$	✓ gradient/ <i>gradiënt</i> CA ✓ substitution/ <i>vervanging</i> CA ✓ equation/ <i>vergelyking</i> CA OR/OF ✓ gradient/ <i>gradiënt</i> CA ✓ substitution/ <i>vervanging</i> CA ✓ equation/ <i>vergelyking</i> CA AO: full marks/ volpunte (3)
1.2	$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(-5 - (-1))^2 + (-2 - (-6))^2}$ $= 4\sqrt{2}$ OR/OF $\approx 5,66$ $SN = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(0 - 2)^2 + (1 - (-1))^2}$ $= 2\sqrt{2}$ OR/OF $\approx 2,83$ $\frac{PQ}{SN} = \frac{4\sqrt{2}}{2\sqrt{2}}$ $= 2$	✓ SF A ✓ value of/ <i>waarde van</i> PQ A ✓ value of / <i>waarde van</i> SN A ✓ $\frac{4\sqrt{2}}{2\sqrt{2}}$ CA (4)
		[11]

QUESTION/VRAAG 2

		
2.1.1	$x^2 + y^2 = r^2$ $(-5)^2 + (12)^2 = r^2$ $r^2 = 169$ $x^2 + y^2 = 169$	✓ substitute/vervang A ✓ equation/vergelyking CA AO: full marks/ volpunte (2)
2.1.2	$m_{OM} = -\frac{12}{5}$ $m_{\text{tang}} = \frac{5}{12}$ $y = mx + c$ OR/OF $y - y_1 = m(x - x_1)$ $12 = \frac{5}{12}(-5) + c$ $y - (12) = \frac{5}{12}(x - (-5))$ $c = \frac{169}{12}$ $\therefore y = \frac{5}{12}x + \frac{169}{12}$ OR/OF $y \cdot y_1 + x \cdot x_1 = r^2$ $y(12) + x(-5) = 169$ $12y - 5x = 169$ $\therefore y = \frac{5}{12}x + \frac{169}{12}$	✓ gradient/gradiënt A ✓ gradient/gradiënt CA ✓ substitution/vervang CA ✓ equation/vergelyking CA OR/OF ✓ F CA ✓ substitution/vervang CA ✓ S CA ✓ equation/vergelyking CA (4)

2.1.3	$L\left(0; \frac{169}{12}\right)$ OR / OF $L(0;14,08)$	✓ x-coordinates/koördinate A ✓ y- coordinates/koördinate CA (2)
2.1.4	$m_{MD} = \frac{12-0}{-5-(-3)} = -6$ $\tan \beta = -6$ Ref/ Verw $\angle = 80,54^\circ$ $\therefore \beta = 99,46^\circ$	✓ gradient/gradient A ✓ SF CA ✓ reference angle/ verw.hk CA ✓ angle/hoek CA (4)
2.2		✓ both x-intercepts/ beide x-afsnitte A ✓ both y-intercepts beide y-afsnitte A ✓ elliptical shape/eliptiese vorm CA (3)
		[15]

QUESTION/VRAAG 3

3.1.1	$\operatorname{cosec} P \times \tan Q$ $= \operatorname{cosec} 119^\circ \times \tan 61^\circ$ $= \frac{1}{\sin 119^\circ} \times \tan 61^\circ$ $\approx 2,06$	✓ substitution/vervanging A ✓ I A ✓ 2,06 CA (3)
3.1.2	$\cos^2(P + 2Q)$ $= \cos^2(119^\circ + 2 \times 61^\circ)$ $\approx 0,24$	✓ substitution/vervanging A ✓ 0,24 CA (2)
3.2	$\frac{1}{2} \tan \theta = 2$ $\tan \theta = 4$ $r^2 = x^2 + y^2$ $r^2 = (4)^2 + (1)^2$ $= \sqrt{17}$ $\sin^2 \theta + \cos^2 \theta = \left(\frac{4}{\sqrt{17}}\right)^2 + \left(\frac{1}{\sqrt{17}}\right)^2$ $= \frac{16}{17} + \frac{1}{17}$ $= \frac{17}{17}$ $= 1$	✓ S A ✓ substitution/vervanging A ✓ r value/waarde van CA ✓ sin ratio/verh CA ✓ cos ratio/verh CA ✓ S CA (6)
3.3	$\sin x = \tan 318^\circ$ $\sin x = -0,9004040443$ Ref / Verw $\angle = 64,21^\circ$ $x = 180^\circ + 64,21^\circ$ or/of $x = 360^\circ - 64,21^\circ$ $x = 244,21^\circ$ or/of $x = 295,79^\circ$	✓ S A ✓ reference angle/verw.hk CA ✓ 244,21° CA ✓ 295,79° CA (4)
		[15]

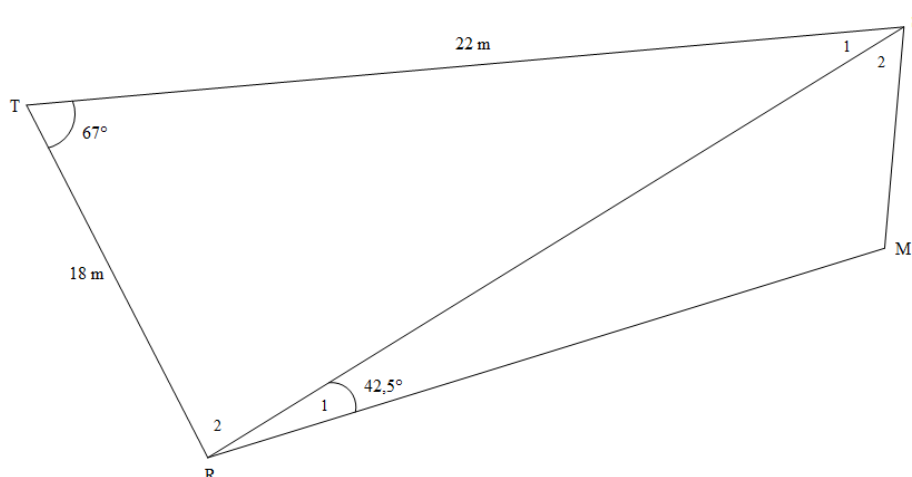
QUESTION/VRAAG 4

4.1.1	$\tan(\pi + A) = \tan A$	✓ $\tan A$	A (1)
4.1.2	$\frac{\tan(\pi + A) \cdot \cos(180^\circ - A) \cdot \sin(360^\circ - A)}{\sin(2\pi + A)}$ $= \frac{\tan A \cdot -\cos A \cdot -\sin A}{\sin A}$ $= \frac{\sin A}{\cos A} \cdot \frac{\cos A}{1}$ $= \sin A$ OR/OF $\frac{\tan(\pi + A) \cdot \cos(180^\circ - A) \cdot \sin(360^\circ - A)}{\sin(2\pi + A)}$ $= \frac{\tan A \cdot -\cos A \cdot -\sin A}{\sin A}$ $= \tan A \cdot \frac{\cos A}{\sin A} \cdot \sin A$ $= \tan A \cdot \cot A \cdot \sin A$ $= \tan A \cdot \frac{1}{\tan A} \cdot \sin A$ $= \sin A$	✓ $-\cos A$ A ✓ $-\sin A$ A ✓ $\sin A$ A ✓ I $\frac{\sin A}{\cos A}$ A ✓ $\sin A$ CA OR / OF ✓ $-\cos A$ A ✓ $-\sin A$ A ✓ $\sin A$ A ✓ I $\frac{\cos A}{\sin A} = \cot A$ A ✓ $\sin A$ CA (5)	
4.2	-1	✓ -1	A (1)
4.3	$\sin x + \cos^2 x \cdot \operatorname{cosec} x = \operatorname{cosec} x$ $\text{LHS} / \text{LK} = \sin x + \cos^2 x \cdot \frac{1}{\sin x}$ $= \frac{\sin^2 x + \cos^2 x}{\sin x}$ $= \frac{1}{\sin x}$ $= \operatorname{cosec} x = \text{RHS} / \text{RK}$	✓ I $\frac{1}{\sin x}$ A ✓ S CA ✓ I $\sin^2 x + \cos^2 x = 1$ A (3)	
			[10]

QUESTION/VRAAG 5

5.1	f: ✓ shape/vorm A ✓ x-intercepts / x-afsnitte A ✓ turning points/draaipunte A (45°; 1), (225°; -1) A ✓ y-intercept / y-afsnit A g: ✓ shape/vorm A ✓ x-intercepts/ x-afsnitte A ✓ turning points/draaipunte A (90°; -2), (270°; 2) A			(7)
5.2	90°	✓ x value/waarde	A	(1)
5.3	360°	✓ period/periode	A	(1)
5.4	$-\frac{1}{2}\cos(x - 45^\circ) = \sin x$ $\cos(x - 45^\circ) = -2\sin x$ A on the graph B on the graph	✓ S A ✓ A on the graph/op die grafiek A ✓ B on the graph/ op die grafiek A		(3)
5.5	$x \in (45^\circ; 225^\circ)$ OR/ OF $45^\circ < x < 225^\circ$	✓ critical values/ kritiese waardes A ✓ notation/ notasie A OR/OF ✓ critical values/kritiese waardes A ✓ notation/notasie A		(2)
				[14]

QUESTION/VRAAG 6

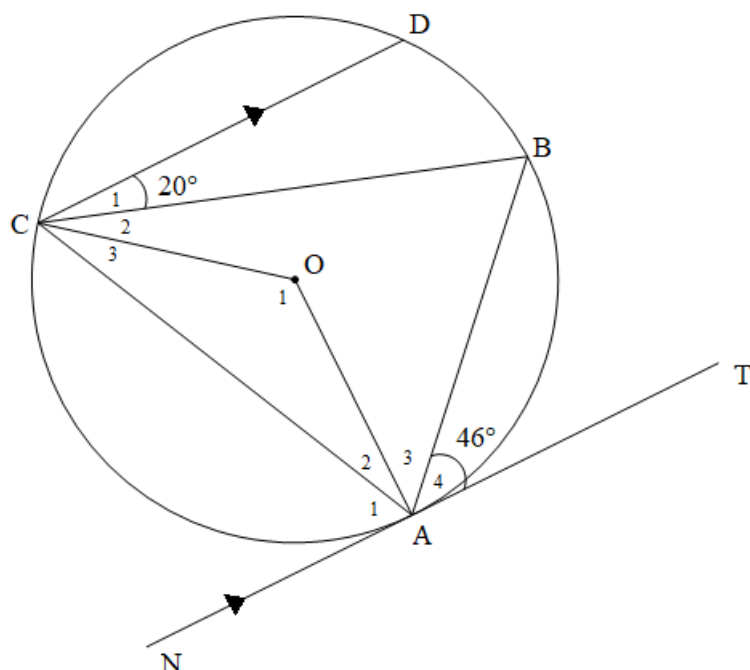


6.1.1	$SR^2 = TS^2 + TR^2 - 2TS \cdot TR \cos T$ $= (22)^2 + (18)^2 - 2(22)(18) \cos 67^\circ$ $= 498,5409462$ $SR \approx 22,33 \text{ m}$	✓ cos rule/reël A ✓ substitution/vervanging A ✓ length of/lengte van SR CA (3)
6.1.2	$\hat{M} = 180^\circ - 67^\circ = 113^\circ$	✓ size of/grootte $\hat{M}$ A (1)
6.2.1	$\frac{SM}{\sin R_1} = \frac{SR}{\sin M}$	✓ sin rule/reël A (1)
6.2.2	$\frac{SM}{\sin 42,5^\circ} = \frac{22,33}{\sin 113^\circ}$ $SM = \frac{22,33 \sin 42,5^\circ}{\sin 113^\circ}$ $= 16,39 \text{ m}$	✓ substitution/verv CA ✓ length of/lengte van SM CA (2)
6.3	$\hat{S}_2 = 24,5^\circ$ $\text{Area of/van } \triangle SMR = \frac{1}{2} SR \times SM \sin \hat{S}_2$ OR/OF $\text{Area of/van } \triangle SMR = \frac{1}{2} m \times r \times \sin \hat{S}_2$ $\text{Area of/van } \triangle SMR = \frac{1}{2} (22,33)(16,39) \sin 24,5^\circ$ $= 75,89 \text{ m}^2$ $\text{Bags/sakke} = \frac{75,89}{15,178} = 5$ 5 bags will be required / sakke sal benodig word	✓ size of $\hat{S}_2$ CA ✓ area rule/reël A ✓ substitution/vervanging CA ✓ area CA ✓ number of bags/aantal sakke CA (5)
		[12]

	$\hat{S}_2 = 180^\circ - 40^\circ = 140^\circ$ $\left(\begin{array}{l} \angle \text{s on a straight line/} \\ \angle \text{e op 'n reguitlyn} \end{array} \right)$	✓ ST	CA
	$\hat{V}_2 = 180^\circ - 40^\circ = 140^\circ$ $\left(\begin{array}{l} \angle \text{s on a straight line/} \\ \angle \text{e op 'n reguitlyn} \end{array} \right)$	✓ ST	CA
	$\hat{S}_2 + \hat{V}_2 \neq 180^\circ$ $\therefore \text{STVP} \left(\begin{array}{l} \text{Not cyclic /} \\ \text{Nie siklies} \end{array} \right) \left(\begin{array}{l} \text{Opp } \angle \text{s not supp /} \\ \text{Teenoorst } \angle \text{e nie supp} \end{array} \right)$ OR / OF	✓ RE	A
	$\hat{V}_2 = \hat{S}_2 = 180^\circ - 40^\circ = 140^\circ$ $\left(\begin{array}{l} \angle \text{s on a straight line/} \\ \angle \text{e op 'n reguitlyn} \end{array} \right)$	✓ ST	CA
	$\hat{V}_1 \neq \hat{S}_2$ OR/OF $\hat{V}_2 \neq \hat{S}_1$	✓ ST	CA
	$\therefore \text{STVP} \left(\begin{array}{l} \text{Not cyclic /} \\ \text{Nie siklies} \end{array} \right) \left(\begin{array}{l} \text{Ext } \angle \neq \text{opp int } \angle / \\ \text{Buite } \angle \neq \text{teenoorst binne } \angle \end{array} \right)$	✓ RE	A
			(3)
			[7]

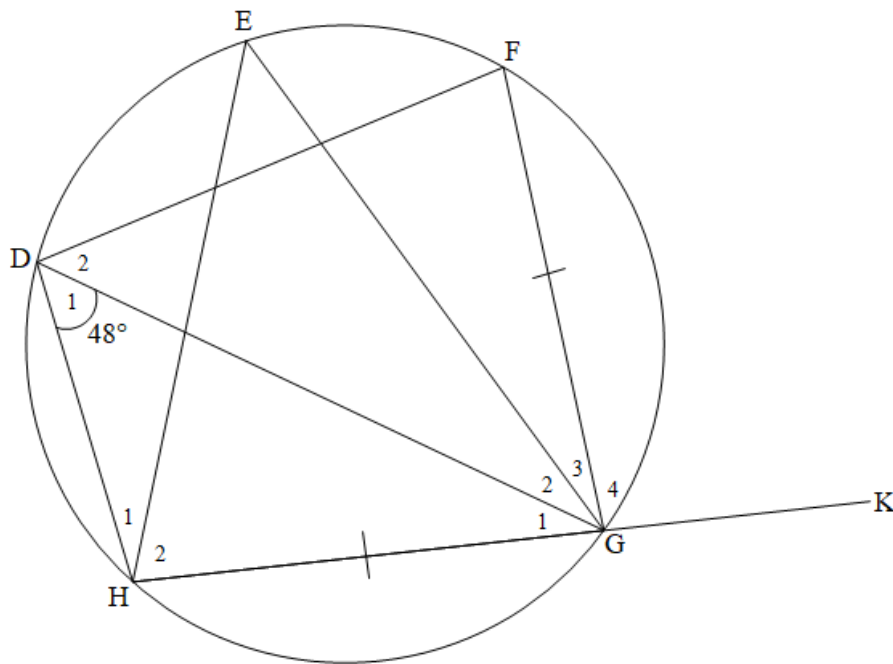
QUESTION/VRAAG 8

8.1

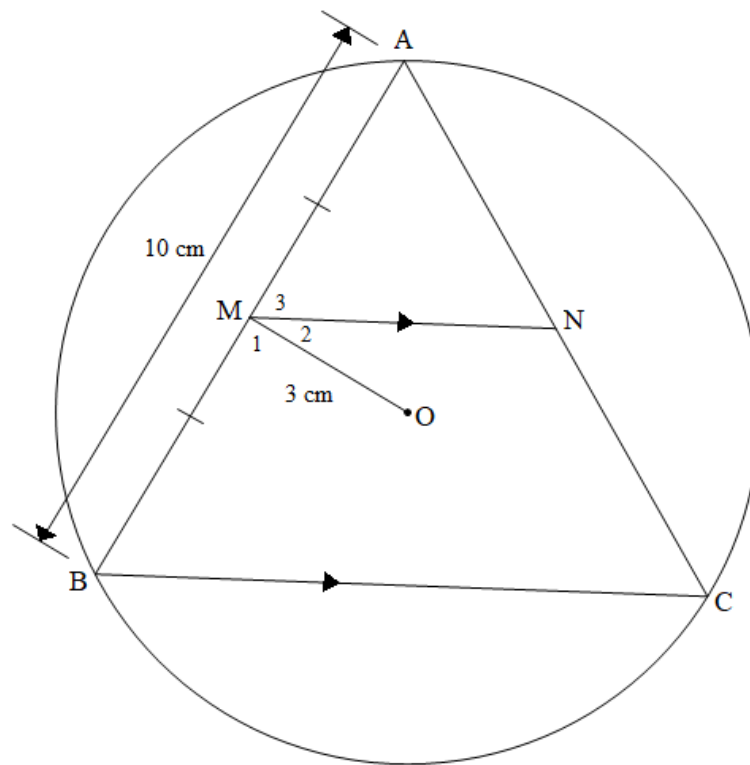


8.1.1	$\hat{BCA} = 46^\circ$ (tan - chord / raaklyn – koord)	✓ ST ✓ RE	A A (2)
8.1.2	$\hat{OAT} = 90^\circ$ $\left(\begin{array}{l} \text{tan} \perp \text{rad} / \\ \text{raaklyn} \perp \text{rad} \end{array} \right)$ $\therefore \hat{A}_3 = 44^\circ$	✓ ST ✓ RE ✓ ST	A A CA (3)
8.1.3	$\hat{A}_1 = 66^\circ$ $\left(\begin{array}{l} \text{alt } \angle\text{s}; CD \parallel NT / \\ \text{verw } \angle\text{e}; CD \parallel NT \end{array} \right)$	✓ ST ✓ RE	CA A (2)
8.1.4	$\hat{B} = 66^\circ$ (tan - chord / raaklyn – koord) $\hat{O}_1 = 132^\circ$ $\left(\begin{array}{l} \angle \text{ at centre} = 2 \times \angle \text{ at circum} / \\ \text{midpts } \angle = 2 \times \text{omtreks } \angle \end{array} \right)$ OR / OF $\hat{A}_2 = 24^\circ$ $\left(\begin{array}{l} \text{tan} \perp \text{rad} / \\ \text{raaklyn} \perp \text{rad} \end{array} \right)$ $\therefore \hat{C}_3 = 24^\circ$ $\left(\begin{array}{l} \angle\text{s opp} = \text{sides} / \\ \angle\text{e teenoor} = \text{sye} \end{array} \right)$ $\therefore \hat{O}_1 = 132^\circ$ $\left(\begin{array}{l} \text{int } \angle\text{s of } \Delta / \\ \text{binne } \angle\text{e van } \Delta \end{array} \right)$	✓ ST ✓ ST ✓ RE ✓ ST ✓ ST ✓ ST	CA CA A OR / OF CA CA CA (3)

8.2

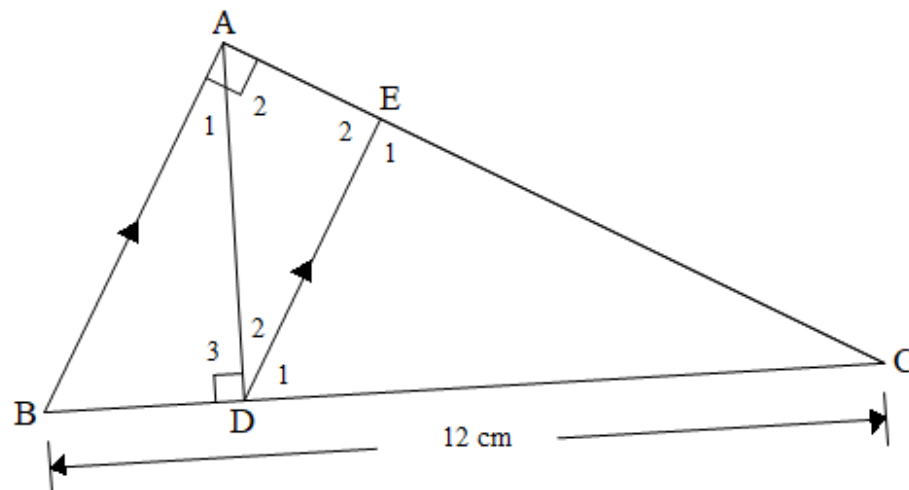


8.2.1	$\hat{E} = 48^\circ$ ($\angle$ s in the same segment / $\angle$ e in dieselfde segment)	✓ ST ✓ RE	A A (2)
8.2.2	$\hat{D}_2 = 48^\circ$ (equal chords / gelyke koorde)	✓ ST ✓ RE	CA A (2)
8.2.3	$\hat{G}_4 = 96^\circ$ (ext $\angle$ of cyclic quad / buite $\angle$ van kdvh)	✓ ST ✓ RE	CA A (2)
			[16]

QUESTION / VRAAG 9


9.1.1 a)	$\hat{M}_1 = 90^\circ$ (line from centre to midpt of chord / lyn vanaf midpt sirkel na midpt vankrd)	✓ ST ✓ RE	A A (2)
9.1.1 b)	$MB = 5 \text{ cm}$ $OB^2 = OM^2 + MB^2$ (Pythagoras) $\therefore OB^2 = 3^2 + 5^2 = 34$ $\therefore OB = \sqrt{34} \approx 5,83 \text{ cm}$	✓ length of/lengte van AB ✓ Pythagoras ✓ length of/lengte van OB	A A CA (3)
9.1.2	$BC = 2MN$ (Midpoint thm / Middelpst) $\therefore BC = 10,24 \text{ cm}$	✓ ST ✓ RE	CA A (2)

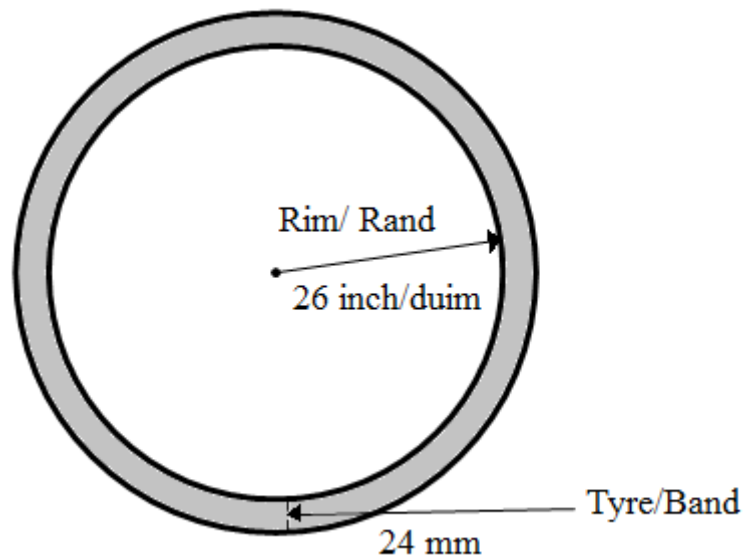
9.2



9.2.1	In $\triangle ADC$ and $\triangle BAC$: $\hat{ADC} = \hat{A} = 90^\circ$ (given/ <i>gegee</i>) $\hat{C}$ is common / <i>gemeen</i> $\therefore \triangle ADC \parallel \triangle BAC$ ($\angle\angle\angle$)	✓ ST ✓ ST ✓ RE	A A A (3)
9.2.2	$\frac{DC}{AC} = \frac{AC}{BC}$ ($\triangle ADC \parallel \triangle BAC$) $\therefore AC^2 = DC \cdot BC$	✓ ST correct ratio / <i>korrekte verh</i>	A (1)
9.2.3 a)	$\frac{DC}{BC} = \frac{CE}{AC}$ (Prop th/<i>ewer st</i>; $DE \parallel AB$)	✓ ST ✓ RE	A A (2)
9.2.3 b)	$\frac{DC}{12} = \frac{2}{3}$ (from/ <i>vanaf</i> 9.2.3a) $\therefore DC = \frac{2}{3} \times 12 = 8 \text{ cm}$	✓ substitution / <i>vervanging</i> ✓ ST	A CA (2)
9.2.3 c)	$\therefore AC^2 = 8 \times 12 = 96 \text{ cm}$ $\therefore AC \approx 9,80 \text{ cm}$	✓ ST ✓ ST	CA CA (2)
			[17]

QUESTION/VRAAG 10

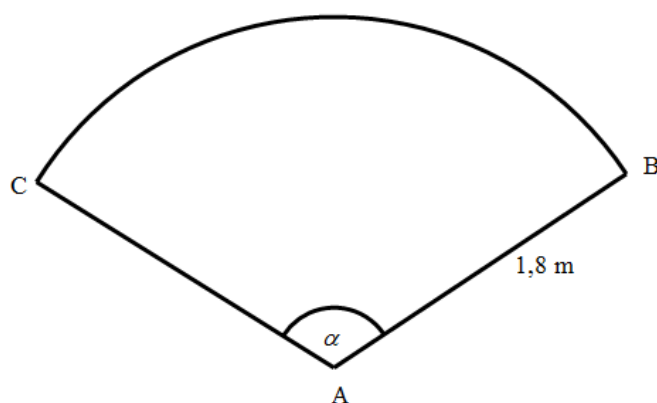
10.1



10.1.1	$26 \text{ inch/ duim} = 26 \text{ inch} / \text{duim} \times \frac{0,0254 \text{ m}}{1 \text{ inch}} \approx 0,66 \text{ m}$	✓answer/antwoord	A (1)
10.1.2	Diameter/ Middellyn $= 0,66 \times 2 + 2 \times 24 \text{ mm} \times \frac{1 \text{ m}}{1000 \text{ mm}} \approx 1,37 \text{ m}$ OR/ OF Radius $= 0,66 + 24 \text{ mm} \times \frac{1 \text{ m}}{1000 \text{ mm}} \approx 0,684 \text{ m}$ Diameter/ Middellyn = $2 \times 0,684 \approx 1,37 \text{ m}$	✓M ✓answer/antwoord OR/OF ✓M ✓answer/antwoord	A CA OR/OF A CA (2)
10.1.3	$v = 60 \text{ km/h} = \frac{60 \text{ km}}{1 \text{ h}} \times \frac{1 \text{ h}}{3600 \text{ s}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 16,67 \text{ m/s}$ $v = \pi D n$ $\therefore 16,67 \text{ m/s} = \pi(1,37)n$ $\therefore n = \frac{16,67}{1,37 \pi}$ $\therefore n \approx 3,87 \text{ rev/s}$ OR/OF	✓conversion/herleiding ✓F ✓SF ✓answer/antwoord OR/OF	A A CA CA OR/OF

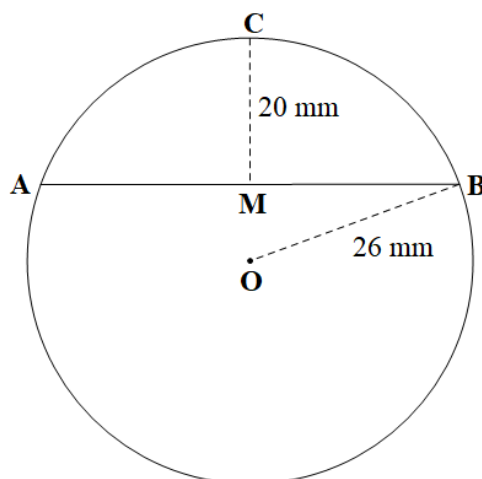
	$v = 60 \text{ km/h} = \frac{60 \text{ km}}{1 \text{ h}} \times \frac{1 \text{ h}}{3600 \text{ s}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 16,67 \text{ m/s}$ $w = \frac{v}{r}$ $\therefore w = \frac{16,67}{0,684} \approx 24,371$ $\therefore w = 2\pi n$ $\therefore n = \frac{24,371}{2\pi} \approx 3,87 \text{ rev/s}$	✓ conversion/herleiding A ✓ F A ✓ SF CA ✓ answer/antwoord CA (4)
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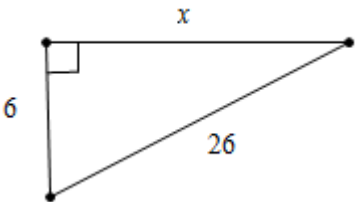
10.2



	Area of sector/van sektor = $\frac{r^2 \theta}{2}$ $2,5 = \frac{(1,8)^2 \alpha}{2}$ $\alpha = \frac{2,5 \times 2}{3,24} = 1,5432 \text{ rad}$ $\therefore \alpha = 1,5432 \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} \approx 88,42^\circ..$ Thus α is an acute angle/Dus is α 'n skerphoek OR/OF Area of a sector/van' n sektor = $\frac{\theta}{360^\circ} \times \pi r^2$ $2,5 = \frac{\alpha}{360^\circ} \times \pi (1,8)^2$ $\alpha = \frac{2,5 \times 360^\circ}{3,24 \pi}$ $\alpha \approx 88,42^\circ$ Thus α is an acute angle/Dus is α 'n skerphoek	✓ F A ✓ SF A ✓ α in rad CA ✓ α in degrees/grade CA ✓ conclusion/gevolgtrekking CA OR / OF ✓ F A ✓ SF A ✓ S CA ✓ α in degrees/grade CA ✓ conclusion/gevolgtrekking CA (5)
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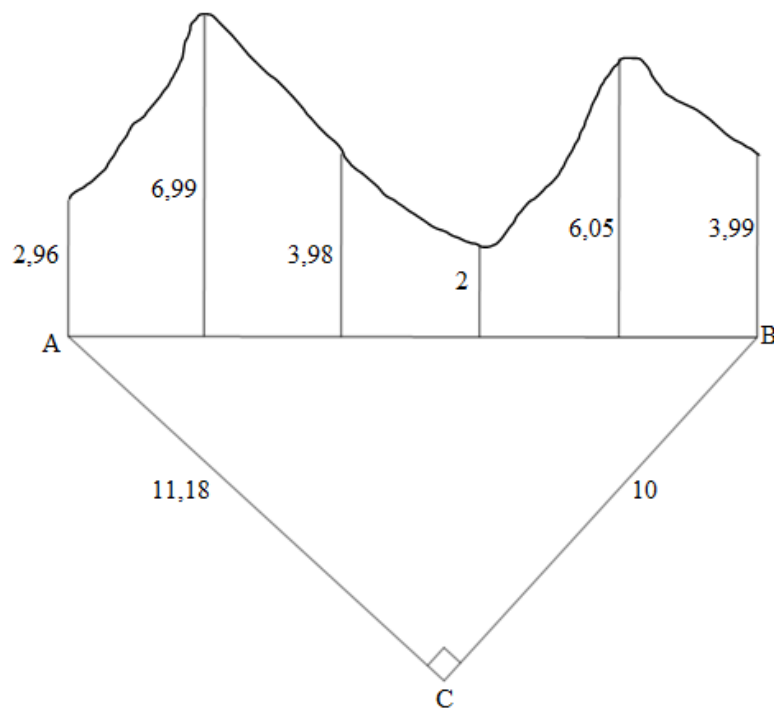
10.3



	$4h^2 - 4dh + x^2 = 0$ $4(20)^2 - 4(52)(20) + x^2 = 0$ $\therefore x^2 = 2560$ $\therefore x \approx 50,6 \text{ mm } \textbf{OR/OF } \therefore = 8\sqrt{10} \text{ mm}$ OR / OF  $x^2 = 26^2 - 6^2 \text{ (Pythagoras)}$ $x^2 = 640$ $\therefore x \approx 25,298...$ $AB = 2x$ $\therefore AB \approx 50,6 \text{ cm } \textbf{OR/OF } = 16\sqrt{10} \text{ mm}$	$\checkmark \textbf{F}$ A $\checkmark$ diameter/middellyn A $\checkmark \textbf{SF}$ CA $\checkmark$ answer/antwoord CA OR / OF $\checkmark 26 - 20 = 6 \text{ mm}$ A $\checkmark$ Pythagoras A $\checkmark$ value of/waarde van x CA $\checkmark$ length of/lengte van AB CA (4)
		[16]

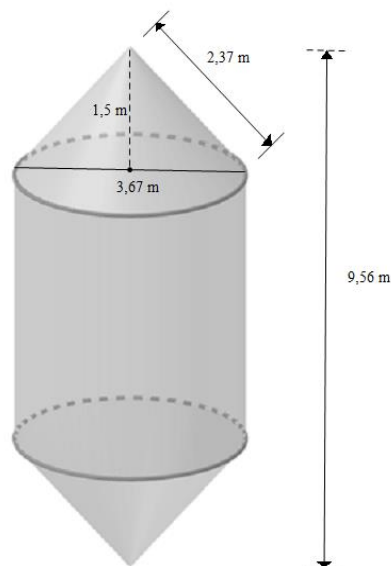
QUESTION / VRAAG 11

11.1



11.1.1	$AB^2 = 11,18^2 + 10^2$ (Pythagoras) $AB \approx 15$ cm	✓ Substitute/vervang ✓ answer/antwoord	A CA (2)
11.1.2	width/ wydte = $\frac{15}{5} = 3$ cm	✓ answer/antwoord	CA (1)
11.1.3	$\text{Area} = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + \dots + o_{n-1} \right)$ $= 3 \left(\frac{2,96 + 3,99}{2} + 6,99 + 3,98 + 2 + 6,05 \right)$ $= 3(22,495)$ $\approx 67,49 \text{ cm}^2$ OR / OF $\text{Area} = a(m_1 + m_2 + m_3 + \dots + m_n)$ $= 3 \left(\frac{2,96 + 6,99}{2} + \frac{6,99 + 3,98}{2} + \frac{3,98 + 2}{2} + \frac{2 + 6,05}{2} + \frac{6,05 + 3,99}{2} \right)$ $= 3(22,495)$ $\approx 67,49 \text{ cm}^2$	✓ F ✓ SF ✓ answer/antwoord OR / OF ✓ F ✓ SF ✓ answer/antwoord	A CA CA A CA CA (3)

11.2



11.2.1 a)	r of cone / van keël = 1,835 m	✓ answer/antwoord	A (1)
11.2.1 b)	height of cylinder / hoogte van silinder = 6,56 m	✓ answer /antwoord	A (1)
11.2.2	Volume of container/ houer $= \pi r^2 h + 2 \times \frac{1}{3} \pi r^2 h$ $= \pi (1,835)^2 (6,56) + 2 \times \frac{1}{3} \pi (1,835)^2 (1,5)$ $\approx 25,46 \pi \text{ m}^3$ OR/OF $\approx 79,97 \text{ m}^3$	✓ F ✓ SF ✓ answer/antwoord	A CA CA (3)
11.2.3	Total surface area / Totale buite opp $= 2\pi r h + 2 \times \pi r \ell$ $= 2\pi (1,835)(6,56) + 2 \times \pi (1,835)(2,37)$ $\approx 32,78\pi \text{ m}^2$ OR/OF $\approx 102,96 \text{ m}^2$ $\therefore$ The material will not be sufficient to cover / Die materiaal sal nie voldoende wees nie	✓ F $2\pi r h$ ✓ F $2 \times \pi r \ell$ ✓ substitution/vervanging ✓ substitution/vervanging ✓ answer/antwoord ✓ conclusion/gevolgtrekking	A A CA CA CA CA (6)
			[17]

TOTAL/TOTAAL:150